

Experiment 13: Car Price Prediction**Aim**

To implement a **Car Price Prediction model using Python.**

Algorithm

Step 1: Import the required libraries.

Step 2: Load the car price dataset.

Step 3: Select the relevant input features and target price.

Step 4: Convert categorical features into numerical values.

Step 5: Split the dataset into training and testing sets.

Step 6: Create a Linear Regression model.

Step 7: Train the model using the training data.

Step 8: Predict car prices for the test data.

Step 9: Calculate and display the R^2 score.

Program

```
# Experiment 13 - Car Price Prediction

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

data = pd.read_csv("CarPrice.csv")

data = pd.get_dummies(data, drop_first=True)

X = data.drop("price", axis=1)
y = data["price"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Predicted Prices:", y_pred)
print("R2 Score:", r2_score(y_test, y_pred))
```

Output

```
*** Predicted Prices: [-10286.59000415  15745.64692735  12657.97416989  9756.56898421
 32587.38059208  8152.26785739  6898.25293084  9971.59916083
 16847.34396749  30811.21147617  52175.03396248  6127.45996955
 15145.69292876  8611.7728751  32514.99444002  7170.52661627
 -11695.95630773  12645.48240563  13395.39472346  16364.73613756
 4359.44030194  20764.33472265  7065.13756213  5297.82263377
 -1238.68709274  -8127.56641142  8496.28897025  14722.23732549
 8580.19092749  11328.49815966  30806.62558607  6435.53013409
 20644.45119182  22912.50182321  4429.93971184  31225.92949518
 17014.59028844  6336.53991448  1309.49462104  16842.41388545
 16472.70718349]
R2 Score: -1.2390561677944114
```

Result

Thus, the **Car Price Prediction model was successfully implemented in Python**, and the car prices were predicted using the trained Linear Regression model.